

Literature Status: Canonical AUS Injective Tensor Products

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Status

This is a lightweight literature-status packet. The target question from Garcia-Lirola and Raja, arXiv:1705.02020, is already answered in a separate later paper by R. M. Causey, arXiv:1705.05484. This packet is not a new proof claim by the run.

Source Question

Source paper: Luis Garcia-Lirola and Matias Raja, *On strong asymptotic uniform smoothness and convexity*, arXiv:1705.02020. The relevant passage is in the introduction, page 2. After recalling known renorming results for injective tensor products, the paper asks whether “the injective tensor product of AUS spaces is an AUS space in its canonical norm.”

The same source paper proves the strong version: Corollary 4.2 says that if X and Y are strongly AUS, then $X \hat{\otimes}_\epsilon Y$ is AUS.

Supporting Answer

Supporting paper: R. M. Causey, *Power type asymptotically uniformly smooth and asymptotically uniformly flat norms*, arXiv:1705.05484. The paper was submitted on 2017-05-15, after the source paper’s 2017-05-04 arXiv submission. Its abstract states the Banach-space consequence: the injective tensor product of two AUS Banach spaces is AUS.

The decisive formal statement is Theorem 1.3. For operators

$$A_0 : X_0 \rightarrow Y_0, \quad A_1 : X_1 \rightarrow Y_1,$$

it says that if both A_0 and A_1 are asymptotically uniformly smooth, then

$$A_0 \otimes A_1 : X_0 \hat{\otimes}_\epsilon X_1 \rightarrow Y_0 \hat{\otimes}_\epsilon Y_1$$

is also asymptotically uniformly smooth. Applying this theorem to the identity operators I_X and I_Y gives the Banach-space statement:

$$X, Y \text{ AUS} \implies X \hat{\otimes}_\epsilon Y \text{ AUS}$$

for the canonical injective tensor norm.

This consequence is also stated explicitly in Corollary 4.5: asymptotic uniform smoothness passes to injective tensor products of Banach spaces.

Why This Is an Exact Literature Answer

The supporting paper explicitly connects itself to the source paper. In the introduction, immediately before Theorem 1.3, Causey says that the tensor-product result extends the recent Garcia-Lirola–Raja result on strong asymptotic uniform smoothness; the bibliography item [13] is the source paper.

Thus the match is exact rather than merely reformulated:

- the source question is about the canonical injective tensor product of plain AUS spaces;
- Causey’s theorem proves precisely that AUS passes to canonical injective tensor products;
- Causey explicitly knew and cited the source strong-AUS theorem as the result being extended.

Scope Limits

This packet does not resolve the source paper’s separate Section 3 remark asking for a reflexive Banach space with an FDD that is AUC but not strongly AUC. It also does not resolve the separate AUC-side question in Remark 4.3 about broader positive classes for injective tensor products of AUC spaces. The resolved question here is only the canonical AUS injective tensor-product question from the introduction.

References

1. Luis Garcia-Lirola and Matias Raja, *On strong asymptotic uniform smoothness and convexity*, arXiv:1705.02020, 2017.
2. R. M. Causey, *Power type asymptotically uniformly smooth and asymptotically uniformly flat norms*, arXiv:1705.05484, 2017.